

Sl. No.	Faculty Name	School / Department	Proposed Case Study / Use Case Title	Problem Statement / Context
1	Dr. Satyanarayan Dhal	SoAS / Physics	AI-Driven Discovery of High-Performance Superhydrophobic/ Supercapacitor Electrodes	Time-consuming and resource-intensive to physically fabricate and test every possible combination of ion beam parameters (e.g., ion type, energy, fluence) to see how it affects supercapacitor performance./ Predicting a material's functional properties (like hydrophobicity) from its core physical/chemical properties is
2	Dr. Swarna Prabha Jena	SoET/ ECE	Lightweight Edge AI-Enabled Autonomous Robotic System for Sustainable Agriculture	A lightweight, low-power, edge AI-enabled robotic platform capable of autonomous navigation, crop health monitoring, precision spraying, weed detection and removal, Leaf diseases and pest Identification and yield estimation — all without dependence
3			Evolutionizing Apparel Production: AI-Enabled Garment Tracking for End-to-End Operational	There is a critical need for an AI-driven garment tracking system that integrates IoT, computer vision, and predictive analytics to ensure end-to-end transparency, quality assurance, and intelligent decision-making from the shop floor to the top floor
4	Dr. Trilochan Rout	SOET/CSE	AI-Enhanced Optical Margin Assessment System for Real-Time Detection of	Breast-conserving surgery , where accurate margin detection is critical to avoid recurrence and maintain cosmetic outcomes.
5	Dr. Sudhansu Kumar Samal	SOET/EEE	AI-driven Smart Electronic Control Units (ECUs) and Vehicle Control Units (VCUs) for the electric vehicle (EV)	Advanced control architecture combining intelligent algorithms with brushless DC motor drives to optimize EV performance and efficiency.
6			Load Side Management for Electrical Sources	Intelligent AI-driven optimization for solar, wind, and generator power distribution based on real-time demand
7	Dr.Abinash Gaya	SOET/ECE	5G ORAN automation and performance analysis using AI	ORAN network automation can lead to cell free MIMO signalling between the network elements and IOT devices. ORAN mimo information must be analyzed using AI to enhance the connectivity
8		SOET/ECE	5G Small cells network KPI management for throughput enhancement using AI	5G small cells design and deployment has significant challenges for local networks with high throughput. study of network parameters can be helpful for better connectivity among IOT devices
9	Dr. Sunil Kumar Mohapatra	SoET/CSE	GenAI-Driven Finger Millet Disease Detection: A Hybrid Deep Learning and Generative Approach for Smart	Existing disease detection systems lack interpretability and fail to provide actionable, personalized insights for effective disease management. Therefore there is a need for a robust and explainable AI

10	Dr Dhawaleswar Rao	SoET/CSE	Development of a Multi-language Code Compiler and Evaluation Platform	With the growing demand for automated coding assessments in academic and competitive programming contexts, there is a lack of platforms that can handle multiple languages securely and efficiently. Manual evaluation is time-consuming and prone to bias. This project focuses on developing an automated, secure, and
11	Laxmikanta Sutar	SoET/CSE	Quantum AI and EDA	How can chiplet-based architectures be used to implement low-power, low-latency, cryogenic classical control electronics for quantum processors , and how does this co-design approach improve quantum gate fidelity, throughput, and scalability compared to conventional room-temperature controllers?
12	Dr. Smitanjali Rout	SoVET/EEE	Artificial Intelligence for Advanced Lithium-Ion Battery Diagnostics and Prognostics	Lithium-ion batteries degrade due to complex aging, reducing performance and reliability in EVs and energy systems. Traditional diagnostic methods struggle to accurately estimate SOC, SOH, faults, and RUL in real conditions, leading to safety risks and
13	Dr. Soumya Jal	SoAHS / Paramedics	Portable Blood Analysis System	Develop a compact LoC device for automated blood analysis (cell count, glucose, lipid profile) using AI image interpretation.
14			Cancer Biomarker Screening Chip	Early detection is critical but constrained by current diagnostic infrastructure. Multi-cancer screening remains a challenge due to high cost and complexity. Emerging microfluidic and AI
15			AI-Based Retinal Imaging and Ophthalmic Screening System	Eye diseases such as diabetic retinopathy, glaucoma, and age-related macular degeneration (AMD) are major causes of preventable blindness. Early detection is critical, yet specialized ophthalmic devices and trained clinicians are limited, especially in rural and low-resource regions. Current diagnostic tools are expensive,
16	Dr. Monalisa Mahapatra	SoPLS	Smart Microfluidics: Harnessing AI for Advanced Cell Culture Drug Screening	Traditional drug screening methods using static 2D cultures are limited in mimicking in vivo conditions, leading to poor predictive accuracy and high drug development costs. This project aims to develop an "AI-integrated smart microfluidic platform" that enables

17	Dr Yashaswi Nayak, Dr. Sujata Chakravarty	SoAS, SoET	SmartAquaria: An AI-IoT Enabled System for Intelligent Detection and Management of Fish Breeding Behaviour	The SmartAquaria system aims to leverage AI and IoT technologies to detect, analyse, and manage fish breeding behaviour automatically. By combining sensor-driven data collection with AI-based behavioural analysis, the system can help fish farmers and aquarium enthusiasts maintain optimal breeding conditions, enhance fish welfare, and improve overall aquaculture efficiency.
18	Dr. Polaki Suman	Biotechnology	AI-Integrated Genomic and Proteomic Framework for Early Detection of Stress and Disease in Shrimp Aquaculture	Shrimp aquaculture forms the backbone of India's Blue Economy, generating substantial export earnings and livelihood opportunities across coastal regions. However, the sector faces recurring challenges from disease outbreaks, heavy-metal contamination, and antibiotic misuse, leading to massive economic losses and environmental degradation. The White Spot Syndrome Virus (WSSV) alone causes up to 80–100% mortality within days of infection, while chronic exposure to copper, cadmium, and lead disrupts shrimp metabolism, immune function, and reproductive health. Current disease surveillance systems rely heavily on post-outbreak diagnosis through centralized laboratory testing, which delays intervention and exacerbates losses. Moreover, the lack of integrated environmental monitoring and real-time pathogen detection tools hampers the ability to predict and prevent outbreaks. The overuse of antibiotics in response to infections further
19	Dr.Prangya Parimita Pradhan	SoET	AI based Solar-Wind power assisted Electric Vehicle charging System (controller design for all converter)	An AI-based Solar-Wind Power Assisted Electric Vehicle Charging System that ensures efficient and reliable energy management using renewable sources. The AI controller will intelligently regulate all converters to optimize power flow, maintain stability, and provide
20 21	Dr. Pushpalatha Dr. Chandra Sekhar Dash/Prof. S.N Padhy	Agriculture SoET	AI based accelerated plant tissue culture and hydroponics Chromite-M SoC Based Edge AI Audio System for Real-Time Speech Synthesis	AI-associated media dispensing to speed up the growth, multiplication and regeneration of plants in vitro , plant growth Most existing language models are cloud-dependent and unsuitable for offline or low-power edge devices. As part of this initiative, an Edge AI accelerator will be developed on the Chromite-M SoC to enable efficient, on-device AI computation. A Small Language Model (SLM) optimized for Indian language processing will be

22	Dr Kamal Kumar Barik	SoET	AI-Integrated Geospatial Technology for Advanced Earth and Coastal Studies	The integration of Artificial Intelligence (AI) with Geospatial Technology has revolutionized Earth and Coastal Studies by enabling intelligent data analysis, pattern recognition, and predictive modeling. AI-driven geospatial analytics can process vast amounts of satellite and sensor data to monitor coastal erosion, land use
23	Dr. Satyabrata Lenka	SoET	Vision-Driven Site Intelligence: Equipment Utilisation & Safety Monitoring	Large sites manage hundreds of heavy assets; idle time adds cost and schedule risk. Worker safety incidents/human-error still prevalent despite existing controls. Manual monitoring and reporting are labour-intensive and reactive.
24	Dr. DJMSNK Sri Veda	CSSAg	AI-Assisted Precision Nutrient and Water Management in Agriculture	Efficient nutrient and water management is a major challenge in agriculture due to spatial and temporal variability in soil and crop conditions. Conventional approaches often lead to overuse or underuse of inputs, affecting both productivity and environmental sustainability. Artificial Intelligence (AI) can be integrated with IoT sensors,
25	Dr. Ayashkanta Jena	SoET-Mechanical	AI-Applications for accurately predicting the surface quality.	AI to forecast the surface quality in CNC machining. The factors affecting surface quality are reviewed and categorized into two key elements—tool center positioning errors and the interaction between the tool edge and workpiece materials. We want a AI model to enable data scarcity
26	Dr. Chinnmayee Dora	SoET	CareerSync: A finetuned LLM based Resume analyzer	The project “CareerSync: A finetuned LLM based Resume analyzer” automates resume evaluation using a Gemini-powered AI. It analyzes resumes and job descriptions to deliver tailored feedback for job aspirants and recruiters. The system provides skill-gap